

Timing Diagram — Periodic and Event-Driven Relationships



Timing Summary

Signal	Peripheral	Prescaler	OCR / Compare Value	Period	Description
Clock second tick	Timer1 (16-bit)	/256 (CS12)	OCR1A = 31249	1.000 s	Real-time clock update
Display mux tick	Timer2 (8-bit)	/64 (CS22)	OCR2 = 249 (0xF9)	2 ms	Digit multiplex, keypad scan
UART communication	USART (async)	—	UBRR = 51 (0x33)	104.17 μ s/bit	9600 baud, 8N1
Heartbeat LED	Timer2 ticks	—	500 $\times$ 2 ms	1.0 s	PD4 toggle (500 ticks)
Edit cursor blink	Timer2 ticks	—	64 $\times$ 2 ms (tick > 6)	128 ms on/off	~3.9 Hz blink rate
Full display frame	Timer2 $\times$ 6 digit	—	6 $\times$ 2 ms	12 ms	~83.3 Hz refresh rate